EFO-KS01	0.02	0.128	2.9	1.28

Fig. 19.3. Top: Romilly (played by David Oyelowo) and Brand (played by Anne Hathaway) discuss Dr. Mann's geological data with him. *Bottom:* One page of data, prepared for the movie by Erika Swanson: the results of chemical analyses of rocks collected on the purported surface of the planet. Several rocks show promising evidence of organic material that could have arisen from living things. *[From* Interstellar*, used courtesy of Warner Bros. Entertainment Inc.]*

In the movie, it turns out that Dr. Mann has faked his data. That's a bit ironic since, of course, Erika faked her data too. She has never made a field trip to Mann's planet. Perhaps someday . . .

In this book I say nothing about the tragedy of Dr. Mann. It's a human tragedy, involving little science. The tragedy's climax is an explosion that severely damages the *Endurance*. The explosion, the damage, and the *Endurance*'s design: that's the stuff of science and engineering, so let's discuss them.

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33 In the movie, when the *Endurance* is in orbit around Mann's planet, we see Gargantua subtending about 0.9 degrees on the sky—nearly twice the size of the Moon as seen from Earth. From this I compute that Mann's planet is about 600 Gargantua radii from the black hole. At this distance, the time required for the planet to travel inward to near Gargantua is at least forty days—a lot longer than the crew *seem* to spend on and near Mann's planet, but reasonable for the outward trip to reach the planet; see Chapter 7.